Part II — Motion and Forces · Chapter 7

Speed, Velocity, and Acceleration

Quantifying how fast objects move, in which direction, and how their motion changes over time

> *A car travels at 60 km/h north. A second car travels at 60 km/h south. They have the same speed but completely different velocities. If the first car brakes from 60 km/h to 0, it accelerates — even though most people would say it 'decelerated.' If a car rounds a curve at constant speed, it is also accelerating — even though its speedometer never changes. This chapter builds the precise vocabulary that resolves all these apparent contradictions.*

7.1 Measuring How Fast Objects Move

The most natural question to ask about a moving object is: how fast is it going? Informally, fast means covering a lot of distance in a short time. Formally, speed is the ratio of distance traveled to the time taken. This ratio — a distance divided by a time — gives a single number that tells us how rapidly position is changing, regardless of direction.

Speed has units of distance per time: meters per second (m/s) in SI, kilometers per hour (km/h) in everyday life, or miles per hour (mph) in some countries. Converting between these is a dimensional analysis problem (Chapter 3). In physics, m/s is the standard unit.

Common Speed	Approximate Value (m/s)
Person walking	1.4 m/s
Cyclist	5–8 m/s
Car on highway	28–33 m/s (100–120 km/h)
Passenger jet	250 m/s (900 km/h)
Sound in air (20°C)	343 m/s
Rifle bullet	900 m/s

Speed of light	3.00×10^8 m/s

Table 7.1 *Representative speeds for comparison. Note the 11 orders of magnitude between walking and light.*

7.2 Average Speed

Average speed is the total distance traveled divided by the total time elapsed. It is a scalar quantity: it has magnitude but no direction. Average speed is always positive (or zero if the object does not move).

$$\text{average speed} = \text{total distance} / \text{total time} = d / \Delta t$$

Average speed uses total distance (not displacement) and total elapsed time.

Example — Average speed calculation A runner completes a 400 m lap in 52.0 s. Average speed = 400 m / 52.0 s = 7.69 m/s. Note: the displacement is 0 (returned to start), but the distance is 400 m. Average speed uses distance.

Example — Multi-segment average speed A car drives 100 km in 1.0 h, stops for 0.5 h, then drives 50 km in 0.5 h. Total distance = 150 km. Total time = 2.0 h. Average speed = 150 km / 2.0 h = 75 km/h.

Common Mistake: Average speed is NOT the average of speeds. If you drive 60 km/h for 1 h and 40 km/h for 2 h, the average speed is (60×1 + 40×2) / (1+2) = 140/3 = 46.7 km/h — not (60+40)/2 = 50 km/h. Always use total distance divided by total time.

7.3 Instantaneous Speed

Average speed tells you how fast an object moved over an interval of time. Instantaneous speed tells you how fast it is moving at one specific moment. A car's speedometer displays instantaneous speed — not average speed.

Formally, instantaneous speed is the limit of average speed as the time interval shrinks to zero. As the time interval Δt approaches zero, the distance Δd also shrinks, but their ratio $\Delta d/\Delta t$ approaches a definite value — the instantaneous speed.

$$\text{instantaneous speed} = \text{limit } (\Delta d / \Delta t) \text{ as } \Delta t \to 0$$

This limit is the derivative of distance with respect to time — a calculus concept. For now, read it as the speed at an instant.

On a position-time graph, the instantaneous speed at any moment equals the magnitude of the slope of the tangent line to the curve at that moment. A tangent line just touches the curve at one point without crossing it. Steep tangent = high speed; shallow tangent = low speed; horizontal tangent = zero speed (momentarily stopped).

Bridge Note *The limit definition of instantaneous speed is the first example of a derivative in this book. Volume I of the advanced series uses derivative notation throughout. The geometric picture of the tangent slope introduced here will make that notation natural when you encounter it.*

7.4 Velocity as Speed with Direction

Velocity is the vector version of speed: it encodes both how fast an object is moving and in which direction. Two objects with the same speed but opposite directions have the same speed but different velocities.

In one dimension, direction is encoded by sign. Positive velocity means motion in the positive direction; negative velocity means motion in the negative direction. In two or three dimensions, velocity is a vector with components along each axis.

Speed	Velocity
Scalar (magnitude only)	Vector (magnitude and direction)
Always ≥ 0	Can be positive, negative, or zero
Tells you how fast	Tells you how fast and which way
Total distance / time	Displacement / time
Never negative	Negative = moving in negative direction
Symbol: v (italic, no arrow)	Symbol: $\vec{v}$ (with arrow) or signed v in 1D

Table 7.2 *Speed versus velocity: a crucial distinction in kinematics.*

Key Point: In one-dimensional motion, we often write velocity as a signed number without an explicit arrow symbol. The sign carries the direction information. A velocity of −15 m/s means 15 m/s in the negative direction — not a speed of −15 (speed cannot be negative).

7.5 Average Velocity

Average velocity is displacement divided by elapsed time. Unlike average speed (which uses distance), average velocity uses displacement and therefore has direction.

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v̄ = Δx / Δt = (x_f − x_i) / (t_f − t_i)
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Average velocity: displacement divided by elapsed time. The bar over v indicates 'average.'

Average velocity can be zero even when the object has been moving: if the displacement is zero (start and finish at the same place), the average velocity is zero. Average speed over the same trip would be positive.

Example — Average velocity vs. average speed A swimmer swims 50 m to the far wall and 50 m back in 60.0 s. Distance = 100 m; average speed = 100/60 = 1.67 m/s. Displacement = 0 m; average velocity = 0/60 = 0 m/s. The swimmer was definitely moving, but the average velocity is zero.

Example — Average velocity with direction A car moves from x_i = +5.0 m to x_f = +65.0 m in 8.0 s. Average velocity = (65.0 − 5.0)/8.0 = +7.5 m/s. The positive sign indicates motion in the positive direction.

7.6 Instantaneous Velocity

Instantaneous velocity is the velocity at one specific moment in time. It is the limit of average velocity as the time interval shrinks to zero. On a position-time graph, instantaneous velocity at any moment equals the slope of the tangent line at that point — including its sign.

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v_inst = limit (Δx / Δt) as Δt → 0
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The slope of the tangent to an x-t curve gives both the magnitude of instantaneous velocity (the steepness) and its direction (the sign of the slope). A positive slope means positive velocity; a negative slope means negative velocity; a zero slope means instantaneous velocity = 0 (the object is momentarily at rest, even if it is about to start moving again).

Example — Instantaneous velocity from a graph At t = 3.0 s on an x-t graph, the tangent line passes through (2.0 s, 4.0 m) and (4.0 s, 12.0 m). Slope = (12.0 − 4.0)/(4.0 − 2.0) = 8.0/2.0 = +4.0 m/s. Instantaneous velocity at t = 3.0 s is +4.0 m/s.

7.7 Positive and Negative Velocity

The sign of velocity tells you the direction of motion relative to the chosen positive direction, not whether the object is speeding up or slowing down. A negative velocity simply means the object is moving in the negative direction.

An object with negative velocity is not necessarily decelerating. An object moving at −10 m/s and then at −15 m/s has increased its speed — it is moving faster in the negative direction. The velocity became ‘more negative,’ which means the speed increased.

Velocity	Direction of Motion	Speed

+12 m/s	Positive direction	12 m/s
−12 m/s	Negative direction	12 m/s
+20 m/s	Positive direction (faster)	20 m/s
−5 m/s	Negative direction (slower)	5 m/s
0 m/s	At rest	0 m/s

Table 7.3 *Velocity sign, direction, and speed — distinguished carefully.*

Common Mistake: Never say 'negative velocity means slowing down.' Negative velocity means moving in the negative direction. Whether the object is speeding up or slowing down depends on whether the magnitude of velocity is increasing or decreasing — not on its sign.

7.8 What Acceleration Means

Acceleration is any change in velocity. This is the single most important conceptual statement in this chapter. Students often think of acceleration as synonymous with 'speeding up,' but this is incomplete and leads to systematic errors.

An object accelerates whenever its velocity changes — in magnitude, in direction, or in both. This means:

- Speeding up in the positive direction: positive acceleration.
- Slowing down in the positive direction: negative acceleration (also called deceleration).
- Speeding up in the negative direction: negative acceleration (the object is already moving negatively, and getting faster at it).
- Slowing down in the negative direction: positive acceleration (the velocity is becoming less negative).
- Turning a corner at constant speed: acceleration (the direction of velocity is changing, so velocity is changing).

Key Point: A car rounding a curve at exactly 60 km/h is accelerating — because the direction of its velocity is changing every moment. Acceleration is any rate of change of velocity, whether that change is in magnitude, direction, or both. This is why acceleration is a vector.

7.9 Average Acceleration

Average acceleration is the change in velocity divided by the elapsed time. It is a vector quantity: it has both magnitude and direction.

$$\bar{a} = \Delta v / \Delta t = (v_f - v_i) / (t_f - t_i)$$

Average acceleration: change in velocity divided by elapsed time. SI unit: m/s².

Example — Average acceleration (speeding up) A car accelerates from $v_i = 0$ to $v_f = +20.0$ m/s in 5.0 s. $\bar{a} = (20.0 - 0)/5.0 = +4.0$ m/s². Positive: velocity increased in the positive direction.

Example — Average acceleration (slowing down) A car brakes from $v_i = +25.0$ m/s to $v_f = +5.0$ m/s in 4.0 s. $\bar{a} = (5.0 - 25.0)/4.0 = -20.0/4.0 = -5.0$ m/s². Negative: velocity decreased (object moving positively, slowing down).

Example — Average acceleration (reversing direction) A ball bounces: $v_i = +8.0$ m/s (upward), then $v_f = -6.0$ m/s (downward) after 1.5 s. $\bar{a} = (-6.0 - 8.0)/1.5 = -14.0/1.5 = -9.3$ m/s².

7.10 Instantaneous Acceleration

Instantaneous acceleration is the acceleration at one specific moment in time: the limit of average acceleration as the time interval approaches zero. On a velocity-time graph, instantaneous acceleration at any moment equals the slope of the tangent to the v-t curve at that point.

$$a_{inst} = \text{limit } (\Delta v / \Delta t) \text{ as } \Delta t \to 0$$

For constant acceleration (the most common situation in this book), average and instantaneous acceleration are identical — the acceleration does not change over time.

7.11 Positive vs. Negative Acceleration

The sign of acceleration tells you the direction of the acceleration vector, which determines how the velocity is changing. To figure out whether an object is speeding up or slowing down, compare the signs of velocity and acceleration:

The Rule: If velocity and acceleration have the SAME sign: the object is speeding up (magnitude of velocity increasing). If they have OPPOSITE signs: the object is slowing down (magnitude of velocity decreasing).

Velocity	Acceleration	Result
Positive (+)	Positive (+)	Speeding up in positive direction
Positive (+)	Negative (–)	Slowing down (moving positively,

		decelerating)
Negative (–)	Negative (–)	Speeding up in negative direction
Negative (–)	Positive (+)	Slowing down (moving negatively, decelerating)
Any	Zero (0)	Constant velocity (no change in speed or direction)
Zero (0)	Non-zero	About to start moving; momentarily at rest

Table 7.4 *Sign of velocity and acceleration combined tells you whether the object speeds up or slows down.*

> **Common Mistake:** Deceleration is NOT a separate physical quantity — it simply means an acceleration directed opposite to the velocity. In SI physics problems, always use the signed acceleration value and determine the effect from comparing signs. Do not introduce the word 'deceleration' in calculations.

7.12 Constant vs. Changing Acceleration

Constant acceleration means the acceleration has the same magnitude and direction throughout the motion. The velocity changes by the same amount in each equal time interval. This is the most important and most common case in introductory physics, and it is the only case for which the kinematic equations of Chapter 8 directly apply.

Changing acceleration means the acceleration itself varies over time. This occurs when the net force on the object changes — for example, in a car where the engine output varies, or in fluid drag where the force depends on speed. Treating changing acceleration with constant-acceleration equations gives wrong answers.

Feature	Constant Acceleration	Changing Acceleration
v-t graph shape	Straight line	Curve
a-t graph shape	Horizontal line	Non-horizontal (varies)
Kinematic equations apply?	Yes	No (need calculus or numerical

		methods)	
Example	Free fall; car on a straight track	Air resistance; variable engine force	

Table 7.5 *Constant versus changing acceleration: a critical distinction for choosing methods.*

> **Key Point:** Free fall — the motion of an object under gravity alone with no air resistance — is constant acceleration. The acceleration is $g = 9.81$ m/s² downward, every moment, for every object regardless of mass. This is one of the most important examples of constant acceleration in this book.

7.13 Velocity-Time Graphs

A velocity-time (v-t) graph plots velocity on the vertical axis and time on the horizontal axis. It is the most powerful tool for analyzing one-dimensional motion because both velocity (read directly from the graph) and acceleration and displacement (read from slope and area) are encoded in it.

Reading a v-t Graph

- The value of v at any time t is read directly from the graph: where the curve is at that moment.
- The slope of the v-t graph = acceleration: steep positive slope = large positive acceleration; negative slope = negative acceleration; flat (slope = 0) = constant velocity.
- The area under the v-t graph (between the curve and the time axis) = displacement over that time interval.
- When the v-t curve crosses $v = 0$, the object has momentarily stopped and is about to reverse direction.
- A v-t graph above $v = 0$ means positive velocity (positive direction); below $v = 0$ means negative velocity.

$$\text{slope of v-t graph} = \Delta v / \Delta t = \text{acceleration} \quad (\text{m/s}^2)$$

$$\text{area under v-t graph} = \text{displacement} \quad (\text{m})$$

v-t Graph Feature	Physical Meaning
Horizontal line at $v = +5$ m/s	Constant positive velocity; zero acceleration
Straight line sloping upward	Constant positive acceleration (speeding up in

	+ direction)
Straight line sloping downward	Constant negative acceleration (slowing down if $v > 0$)
Curve (changing slope)	Non-constant acceleration
Crosses $v = 0$ from above	Object stopped momentarily; reverses to negative direction
Large positive area under curve	Large positive displacement
Area below t-axis ($v < 0$)	Negative displacement (moved in negative direction)

Table 7.6 *Features of a v-t graph and their physical meanings.*

Example — Area under v-t graph A car accelerates from rest to 20 m/s over 5.0 s at constant acceleration. The v-t graph is a triangle with base 5.0 s and height 20 m/s. Area = ½ × 5.0 × 20 = 50 m. The car displaced 50 m during the acceleration.

7.14 Acceleration-Time Graphs

An acceleration-time (a-t) graph plots acceleration on the vertical axis and time on the horizontal axis. For constant acceleration — the most common case in this book — the a-t graph is simply a horizontal line.

- A horizontal line above $a = 0$: constant positive acceleration (object speeding up in positive direction or slowing down in negative direction).
- A horizontal line below $a = 0$: constant negative acceleration.
- A line at $a = 0$: zero acceleration; constant velocity.
- The area under an a-t graph equals the change in velocity over that time interval: $\Delta v = a\,\Delta t$.

$$\text{area under a-t graph} = \Delta v = \text{change in velocity} \quad (\text{m/s})$$

The a-t, v-t, and x-t graphs form a hierarchy: the slope of the x-t graph gives v; the slope of the v-t graph gives a. Reading down this hierarchy involves taking slopes; reading upward involves computing areas. Keeping this three-graph structure in mind will support your understanding through Chapter 9.

Graph	Slope Gives	Area Gives
x-t (position-time)	Velocity (m/s)	(not directly used)

v-t (velocity-time)	Acceleration (m/s²)	Displacement (m)
a-t (acceleration-time)	(rate of change of accel.)	Change in velocity (m/s)

Table 7.7 *The slope-area relationships across the three kinematic graphs.*

7.15 Common Misconceptions About Acceleration

Misconception 1: Acceleration means speeding up

Acceleration is any change in velocity — including slowing down, changing direction, or reversing. A car braking from 30 m/s to 0 m/s is decelerating, which is a form of acceleration. Circular motion at constant speed involves continuous acceleration.

Misconception 2: Zero velocity means zero acceleration

An object can have zero velocity and non-zero acceleration simultaneously. A ball thrown straight up has $v = 0$ at the peak of its arc, but its acceleration is -9.81 m/s² (downward) at that same instant. The acceleration never disappears; only the velocity is momentarily zero.

> **Common Mistake:** At the peak of a thrown ball's trajectory: $v = 0$, but $a = -9.81$ m/s². The acceleration is not zero at the peak. If it were, the ball would hang in the air forever.

Misconception 3: Negative acceleration always means slowing down

Negative acceleration means the acceleration vector points in the negative direction. If the object is also moving in the negative direction (negative velocity), negative acceleration means it is speeding up — in the negative direction. Compare signs of v and a to determine whether the object is speeding up or slowing down.

Misconception 4: Acceleration and velocity always point in the same direction

Acceleration and velocity can point in opposite directions. When they do, the object slows down. When they point in the same direction, the object speeds up. This is why a ball thrown upward slows down (velocity upward, acceleration downward) and why a ball falling after the peak speeds up (both velocity and acceleration downward).

Misconception 5: A steep v-t graph means large displacement

A steep slope on a v-t graph means large acceleration, not large displacement. Displacement is the area under the v-t curve. A flat line at high velocity produces more displacement per unit time than a steep slope starting near zero.

Chapter Summary

Speed and velocity both describe how rapidly position changes, but velocity also encodes direction. Acceleration is any change in velocity — in magnitude, direction, or both.

- Average speed = total distance / total time (scalar, always ≥ 0).
- Average velocity = displacement / elapsed time (vector, can be negative or zero).
- Instantaneous speed/velocity = the limit of average speed/velocity as $\Delta t \to 0$; equals the slope of the tangent to the x-t curve.
- Average acceleration = $\Delta v / \Delta t$ (vector); SI unit m/s^2.
- If v and a have the same sign: object speeds up. If opposite signs: object slows down.
- Constant acceleration: v-t graph is a straight line; a-t graph is a horizontal line.
- v-t graph: slope = acceleration; area under curve = displacement.
- a-t graph: horizontal line for constant acceleration; area = change in velocity.
- Acceleration ≠ speeding up. Zero velocity ≠ zero acceleration. Negative acceleration ≠ slowing down (always compare signs of v and a).

Key Terms

speed: The scalar magnitude of velocity; total distance divided by total time; always ≥ 0.

average speed: Total distance traveled divided by total elapsed time; a scalar quantity.

instantaneous speed: The speed at one specific instant; the magnitude of the instantaneous velocity; equals the magnitude of the slope of the tangent to the x-t graph.

velocity: A vector quantity describing the rate of change of position; displacement divided by time; includes both magnitude (speed) and direction.

average velocity: Displacement divided by elapsed time: $\bar{v} = \Delta x / \Delta t$; a vector quantity.

instantaneous velocity: Velocity at a specific instant; the limit of $\Delta x/\Delta t$ as $\Delta t \to 0$; equals the slope of the tangent to the x-t graph (including sign).

acceleration: The rate of change of velocity: $a = \Delta v / \Delta t$; a vector quantity; any change in velocity (magnitude or direction).

average acceleration: Change in velocity divided by elapsed time: $\bar{a} = \Delta v / \Delta t$; SI unit

m/s^2.

instantaneous acceleration: Acceleration at a specific instant; equals the slope of the tangent to the v-t graph.

constant acceleration: Acceleration that does not change over time; produces a straight line on a v-t graph and a horizontal line on an a-t graph.

deceleration: Informal term for acceleration directed opposite to the velocity (the object is slowing down); not a separate physical quantity from acceleration.

free fall: Motion under gravity alone, with no air resistance; produces constant downward acceleration of $g = 9.81\ m/s^2$.

velocity-time graph: A graph of velocity vs. time; slope = acceleration; area under curve = displacement.

acceleration-time graph: A graph of acceleration vs. time; horizontal line for constant acceleration; area = change in velocity.

Concept Review Questions

Answer in complete sentences. No calculation required.

1. What is the difference between average speed and average velocity? Give a situation where they are equal and one where they are different.
2. Can an object have zero average velocity but non-zero average speed? Explain with an example.
3. A car travels north at 40 km/h and another travels south at 40 km/h. Do they have the same speed? The same velocity? Explain.
4. What does it mean for an object to accelerate? Give three physically different examples of acceleration.
5. A ball is thrown straight up. At the peak of its trajectory, what is its velocity? What is its acceleration? Are either of these zero?
6. An object moves with velocity -12 m/s and acceleration $-3\ m/s^2$. Is it speeding up or slowing down? Explain.
7. What does the slope of a v-t graph represent? What does the area under a v-t graph represent?
8. Why does a car rounding a curve at constant speed have acceleration? What direction does the acceleration point?

9. Explain the difference between instantaneous velocity and average velocity. Under what conditions are they equal?
10. A v-t graph shows a horizontal line at $v = -6$ m/s for 5 seconds. What is the acceleration? What is the displacement over those 5 seconds?

Speed and Velocity Problems

Show all work. Report speed as positive; report velocity with sign.

11. A sprinter runs 100 m in 9.85 s. What is the average speed in m/s? In km/h?
12. A car travels from city A to city B (240 km apart) in 2.5 h. Then returns in 3.0 h. (a) Average speed for each leg. (b) Average velocity for the whole trip. (c) Average speed for the whole trip.
13. An object starts at $x = +8.0$ m and moves to $x = -6.0$ m in 7.0 s. (a) What is the displacement? (b) What is the average velocity? (c) What is the distance traveled (assume no reversal)?
14. A swimmer swims 50.0 m at 1.20 m/s, turns, and swims 50.0 m back at 1.50 m/s. (a) Time for each leg. (b) Total distance. (c) Total displacement. (d) Average speed for the round trip. (e) Average velocity for the round trip.
15. An x-t graph has a straight line from (0 s, −3.0 m) to (6.0 s, +9.0 m). Find the average velocity over this interval.
16. At $t = 2.0$ s, a car is at $x = +10.0$ m. At $t = 5.0$ s, it is at $x = +25.0$ m. Find the average velocity.

Acceleration Problems

17. A car accelerates from rest to 28.0 m/s in 8.0 s. Find the average acceleration.
18. A bus brakes from 72.0 km/h to 0 in 6.0 s. Find the average acceleration in m/s^2.
19. A ball thrown upward has $v_i = +15.0$ m/s and reaches $v_f = 0$ m/s at the peak after 1.53 s. (a) Find the average acceleration. (b) What is its direction? (c) Is this consistent with free fall ($g = 9.81\ m/s^2$)? Why?
20. An object has $v_i = -10.0$ m/s and $v_f = -25.0$ m/s after 5.0 s. (a) Find the acceleration. (b) Is the object speeding up or slowing down? How do you know?

21. A train's velocity changes from +20.0 m/s to −5.0 m/s in 10.0 s. Find the average acceleration.
22. An object has $a = +3.0$ m/s². Starting from $v_i = +2.0$ m/s, what is the velocity after 4.0 s?

Motion Graph Interpretation Exercises

Use the following v-t graph description: the graph has three segments. Segment A: v increases linearly from 0 to +12 m/s over 0–4 s. Segment B: v stays constant at +12 m/s from 4–7 s. Segment C: v decreases linearly from +12 m/s to 0 over 7–10 s.

23. Find the acceleration during each segment (A, B, C).
24. Find the displacement during each segment.
25. Find the total displacement from $t = 0$ to $t = 10$ s.
26. Sketch the corresponding a-t graph for 0–10 s.
27. Sketch the corresponding x-t graph for 0–10 s (assume $x = 0$ at $t = 0$). What shape is each segment?
28. A v-t graph shows a straight line from (0 s, +8 m/s) to (5 s, −2 m/s). (a) Find the acceleration. (b) At what time does the object momentarily stop? (c) Find the displacement from $t = 0$ to $t = 5$ s.

Mixed Kinematics Practice

29. A car is at rest. It accelerates at +4.0 m/s² for 5.0 s. (a) What is its final velocity? (b) What is the area under the v-t graph for this interval? (c) What does that area represent?
30. A ball rolls with $v_i = +6.0$ m/s and $a = -2.0$ m/s² (friction). (a) When does it stop? (b) What is its velocity at $t = 1.0$ s, $t = 2.0$ s, $t = 4.0$ s? (c) Is it still decelerating at $t = 4.0$ s? Explain.
31. Two cars start from the same point at the same time. Car A has $v = +15$ m/s constant. Car B starts at rest and has $a = +3.0$ m/s². (a) After how many seconds does Car B reach the same speed as Car A? (b) Draw rough v-t graphs for both on the same axes. (c) Which car has traveled farther at $t = 5.0$ s?

32. Sketch an x-t, v-t, and a-t graph for: 'A ball is dropped from rest and falls for 3 seconds under constant gravitational acceleration (downward = negative).'

Chapter 7 Checkpoint

I can...	Section
Calculate average speed using total distance / total time	7.2
Calculate average velocity using displacement / elapsed time	7.5
Explain the difference between speed and velocity with an example	7.4
Calculate average acceleration using Δv / Δt	7.9
Determine if an object is speeding up or slowing down from signs of v and a	7.11
Read slope and area from a v-t graph and state what each represents physically	7.13
Read slope and area from an a-t graph	7.14
Explain why zero velocity does not imply zero acceleration	7.15
Explain why negative acceleration does not mean slowing down	7.11, 7.15
Identify all five common misconceptions about acceleration	7.15

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